

CodeTrain Instructor Handbook

Guidance for Richard & Joshua delivering 16-week curriculum

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1 Welcome

This handbook supports the Codetrain instructors in delivering the 16-week CodeTrain AI curriculum to 65+ students across weekday, weekend, and online cohorts. It contains:

- Common student struggles per phase
- Recommended interventions
- Demo Day logistics
- Cohort-specific adaptations
- Ghana-context examples

2 Phase 1: Foundations (Weeks 1-4)

2.1 Learning Goals

- Students transition from JavaScript to Pythonic thinking
- Master variables, functions, scope, OOP
- Complete Gate 1: Python mastery

2.2 Common Struggles

“What’s the difference between `x = [1, 2]` and `x = (1, 2)`?”

- Students confuse list (mutable) vs tuple (immutable)
- **Intervention:** Use street analogy: “List is like a mutable marketplace stall (you can rearrange goods); tuple is like an immutable store sign (you can’t change it)”
- Have them predict: “Will this code work?” before running

“I don’t understand scope (LEGB rule)”

- JavaScript has function scope; Python adds enclosing scope (tricky for JS devs)
- **Intervention:** Draw boxes: Global box → Enclosing box → Local box. Visual helps.
- Use the notebook’s LEGB examples; don’t rush.

“List comprehensions feel like magic”

- Students write loops instead: `result = []; for x in items: result.append(x*2)`
- **Intervention:** Show them the 1-to-1 mapping: `[x*2 for x in items]` = loop without intermediate list
- Practice: Give them 5 loops; ask them to rewrite as comprehensions

2.3 Session Structure

1. **Live code-along** (50 min): Demonstrate the concept, type slowly, make mistakes
2. **Guided practice** (20 min): Students code while you watch (share screen on weekday)
3. **Exercise time** (10 min): Assign 2–3 exercises; walk around or monitor chat

2.4 Ghana Context to Inject

- Kwame, Ama, Yaw, Akosua as character names (familiar)
- MTN subscriber manager, MoMo transactions, Accra market data
- “Store receipts” analogy for dictionaries (list of (item, price) tuples)

2.5 Pacing Notes

- **Tight timeline:** 4 weeks for 3 notebooks + exercises
- If stuck on scope: assign extra Scope Patrol exercises (in notebook 2)
- Do NOT skip list comprehensions (it’s needed for Pandas)

2.6 Gate 1 Check (End of Week 4)

Watch for: “Do students write `[x for x in items if x > 5]` naturally?”

- If yes → Pass Gate 1
- If no → Assign extra comprehension drills before Mini Demo Day 1

3 Phase 2: Analytics (Weeks 5-8)

3.1 Learning Goals

- Load, explore, clean, visualize data
- Master `.loc[]` and `.iloc[]` indexing
- Complete Gate 2: Pandas fluency

3.2 Common Struggles

“.loc[] vs .iloc[] - which one do I use?”

- **Intervention:** Create a cheat sheet they can print:
 - `.loc[]` = label (column name, like dictionary key)
 - `.iloc[]` = integer position (0, 1, 2...)
- Have them predict: “Will `df.loc[0]` get the first row?” (No! 0 is a label, not position)

“My dataset has 1M rows - Pandas is slow”

- Real issue, not their fault
- **Intervention:** Teach `.sample(10000)` to work with subset; use `eval: false` for large operations

“Visualization overwhelm - so many plot types”

- **Intervention:** Decision tree: “Is it time series? → Line plot. Two variables? → Scatter. Distribution? → Histogram.”
- Focus on 3 types: line, scatter, histogram (bar comes later)

3.3 Ghana Data Examples

- Use `house_rentals.csv` throughout (real Ghana housing data)
- Discuss real price ranges in Accra/Kumasi
- Explain why cleaning matters: “Bad data → bad predictions → bad decisions for homebuyers”

3.4 Gate 2 Check (End of Week 8)

Students should:

- ☐ Use `.loc[mask, col]` without syntax lookup
- ☐ Answer “How many transactions per user?” with `groupby().size()`
- ☐ Create 3-panel visualizations (subplots)

4 Phase 3: Intelligence (Weeks 9-12)

4.1 Learning Goals

- Train, evaluate, interpret ML models
- Understand precision vs recall trade-offs

- Complete Gate 3: Model building competency

4.2 Common Struggles

“My train/test split leads to different results each time”

- They didn’t set `random_state`
- **Intervention:** “Always set `random_state=42` for reproducibility. It’s not magic—it just freezes randomness.”

“Random Forest has 1000 parameters - how do I know what to change?”

- Overwhelm is real
- **Intervention:** Teach the “Big 3”: `n_estimators` (more trees = slower but better), `max_depth` (control overfitting), `random_state`
- Leave GridSearchCV for later cohorts if they struggle

“I’m getting 95% accuracy but my teacher says it’s wrong”

- Likely data leakage (fit on test data) or class imbalance
- **Intervention:** Ask: “Did you `.fit()` before seeing test data?” and “Are 95% of your samples one class?”

4.3 Ghana Context

- Ghana housing prices: show feature importance (bedrooms >> color)
- MoMo fraud detection: explain precision vs recall (false positives cost customer trust)
- Cocoa yield prediction: show how ML helps smallholders

5 Phase 4: Advanced AI (Weeks 13-16)

5.1 Learning Goals

- Build neural networks in PyTorch
- Understand transfer learning
- Complete capstone project

5.2 GPU/Colab Setup

- **DO:** Send Colab link before Week 13 (students need time to set up)
- **DO:** Require students to test Colab access in Week 12
- **DON’T:** Assume they know how to use Colab (show them step-by-step)
- Backup: Provide pre-trained model checkpoints for offline work

5.3 Common Struggles

“GPU is out of memory”

- Likely: batch size too large or model too big
- **Intervention:** Start with small batch (32), small model (2 conv layers)

“My loss isn’t decreasing”

- Learning rate wrong, or model not seeing data
- **Intervention:** Try `lr = 0.001` first (very conservative). Print batch loss every 10 iterations.

“Transfer learning - do I freeze weights?”

- Yes, at first. Then unfreeze for fine-tuning.
- **Intervention:** Provide their notebook with boilerplate code (don’t make them write from scratch)

6 Demo Day Logistics

6.1 Mini Demo Days 1-3 (15 min per student)

Setup:

- 10 min: Student presents
- 5 min: Q&A
- Score in real-time using rubric

Student Prep:

- Send project link 24 hrs before
- Download/test their code
- Note any issues to ask about

Your Role:

- Ask 2–3 probing questions (not gotcha questions)
- Praise effort; be constructive on feedback
- Take a screenshot of top 3 projects to share with next cohort

6.2 Final Demo Day (20 min per team)

Setup:

- 15 min: Team presents

- 5 min: Q&A
- Use full rubric (50 points)

Team Dynamics:

- Ask each member one question individually
- Score each person separately (fairness)
- Note any free-rider behavior for feedback

7 Cohort-Specific Guidance

7.1 Weekday Cohort (Mon/Wed/Fri afternoons)

- **Audience:** Working professionals, tired after work
- **Pacing:** Slightly slower (give breaks)
- **Homework:** Expect less time investment (2 hrs/wk max)
- **Tip:** Record sessions; students replay at own pace

7.2 Weekend Cohort (Saturday mornings)

- **Audience:** Energetic; dedicated time
- **Pacing:** Can go deeper; 4-hour sessions allow immersion
- **Homework:** Weekend coding projects (Capstone can be more ambitious)
- **Tip:** Use Sunday office hours to catch up stragglers

7.3 Online Cohort (Async + live Q&A)

- **Audience:** Distributed, variable time zones
- **Pacing:** Self-paced videos; uniform live Q&A
- **Homework:** Video submissions (code-along recordings)
- **Tip:** Use office hours to build community (many feel isolated)

8 Office Hours & 1:1 Support

8.1 When to Escalate to 1:1

- Student scores <15 on a Demo Day
- Student is “stuck” (can’t do loop/function/indexing)
- Student missed 2+ weeks
- Student is disengaged (quiet, not submitting)

8.2 1:1 Session Structure (30 min)

1. **Diagnose** (5 min): “What’s the blocker?”
2. **Teach** (15 min): Whiteboard; walk through concept
3. **Practice** (8 min): Have them code while you watch
4. **Plan** (2 min): What to practice before next class?

8.3 Resources to Offer

- Links to tutorials (Real Python, freeCodeCamp)
- Related exercises from their notebook
- Peer study buddy matching

9 Ghana Context Tips

9.1 Cultural Relevance

- Use student names: Kwame, Ama, Yaw, Akosua, Kofi, Abena
- Reference Accra, Kumasi, Tema, Cape Coast (students’ hometowns)
- Use GHS (Ghana Cedis) for all monetary examples
- Reference MTN, mPharma, Hubtel (local companies students know)

9.2 Problem Framing

- “Why is this relevant?” before every unit
- Connect to local problems: crop disease, student retention, financial inclusion
- Celebrate when students solve problems that matter in Ghana

9.3 Capstone Ideas (Student Suggestions)

- Cocoa yield prediction (help farmers)
- Student dropout early warning (keep students in school)
- Mobile money fraud detection (trust in mPesa/MTN)
- Electricity load forecasting (help Accra avoid blackouts)
- Crop disease detection (smartphone app for farmers)

10 FAQs (Answers for Your Use)

Q: A student hasn’t done anything for 2 weeks. What do I do?

- Reach out 1:1: “Haven’t seen you – everything OK?”
- Offer to catch up synchronously
- Check if it’s technical (no internet) or personal (life event)

- Celebrate if they return

Q: Can I change the pacing for my cohort?

- Yes! Use Delivery Schedule (PDF) as a template
- If students are ahead: add stretch projects (deploy to web, submit to Kaggle)
- If students are behind: extend phase by 1 week, compress next phase

Q: What if we don't have enough time for all 4 phases?

- Prioritize Phase 2 (Pandas) – it's the foundation for everything
- Compress Phase 4 – focus on one deep learning project vs multiple exercises
- Students can self-study Phase 4 after course ends

Q: How do I know if a student is actually learning?

- Can they explain concepts in their own words (not copy-paste)?
- Do they ask good debugging questions?
- Can they solve a variant of an exercise (not just replicate)?

11 Your Success as an Instructor

You are successful when:

- • >80% pass Gate 1 (Python mastery, Week 4)
- • >80% pass Gate 2 (Pandas fluency, Week 8)
- • >80% pass Gate 3 (Model building, Week 12)
- • >70% complete capstone with portfolio-ready code
- • Students can explain their code to a non-technical friend
- • Students teach each other (peer learning happens)

12 Before You Start

- ☐ Read Curriculum Strategy (PDF) (understand the big picture)
- ☐ Read Delivery Schedule (PDF) (your week-by-week roadmap)
- ☐ Review all exercise notebooks (you should solve them first!)
- ☐ Set up your Colab + test GPU access
- ☐ Create a Slack/Discord for student Q&A



Print rubrics and post in office

13 During the 16 Weeks



Give feedback on exercises within 48 hours



Celebrate Demo Days (take photos, share with future cohorts)



Adjust pacing weekly based on student progress



Collect 1-2 student stories of “aha!” moments to motivate future cohorts

14 After the 16 Weeks



Collect student feedback (what worked, what didn't)



Identify students for mentorship/jobs at CodeTrain



Archive top capstone projects as examples



Share lessons learned with Ricky for next cohort curriculum improvements

You've got this. Teaching Python to MERN developers is unique—you're their bridge to a new domain. Lean on the “why” (the notebooks) and the “how” (demonstrations). Good luck!